

RW-LoRA: Random-Walk-Based LoRA Fine-Tuning Scheme

Xingran Chen, Rohit Bhagat, Ghadir Ayache, Rawad Bitar, Yanmin Gong, and Salim El Rouayheb

Abstract—Parameter-efficient fine-tuning methods such as LoRA have become a standard approach for adapting large foundation models. However, existing federated or gossip-based LoRA methods either rely on centralized aggregation or require repeated synchronization among multiple model copies, leading to significant communication overhead. In this paper, we propose a random-walk-based LoRA fine-tuning scheme, i.e., RW-LoRA algorithm. Instead of maintaining multiple model replicas, a single model token traverses the network and is updated sequentially using local fine-tuning objectives. This design eliminates the need for global synchronization and substantially reduces communication and computation costs. We provide rigorous convergence guarantees for non-convex objectives under standard assumptions. Empirical results on multiple NLP tasks and graph topologies show that the proposed method achieves competitive task performance while requiring substantially less communication and computation than gossip-based LoRA. These results suggest that random-walk-based LoRA is a promising approach for resource-efficient fine-tuning of large models in decentralized environments.

I. INTRODUCTION

Foundation models such as GPT-4 [1], LLaMA [2], and BERT [3] have transformed artificial intelligence, achieving strong performance on tasks ranging from translation to summarization [4], [5]. Their flexibility comes largely from their ability to be adapted to domain-specific data, but their scale – often 10^8 to 10^{11} parameters – makes full fine-tuning computationally expensive, communication-heavy, and prone to overfitting. As a result, parameter-efficient fine-tuning (PEFT) methods, especially low-rank adaptation (LoRA), have become increasingly important because they greatly reduce the number of trainable parameters and lower the computational cost of fine-tuning [6]. This advantage is particularly important in distributed and resource-constrained applications, such as healthcare, edge personalization, and enterprise AI. In these settings, privacy, bandwidth, and data-sovereignty constraints often prevent raw data from being centralized, so fine-tuning must be carried out locally or through distributed collaboration. Full-parameter fine-tuning in such environments is often impractical due to the large computational burden and the high communication cost of exchanging model updates, while PEFT

substantially reduces both communication and computational overhead, making it a natural fit for distributed fine-tuning under resource constraints. We consider LoRA-based PEFT.

To enable LoRA in distributed environments, several distributed LoRA fine-tuning frameworks have been proposed. Recent federated LoRA methods, such as FLoRA [7], FedALT [8], FedLoRA [9], FRLoRA [10], FedSA-LoRA [11], FedEx-LoRA [12], and FedMomentum [13], address issues such as heterogeneous rank adaptation, but they still rely on centralized aggregation through a parameter server, which introduces substantial communication and memory overhead as well as a single point of failure. Recent concurrent work [14] removes central coordination via gossip-based aggregation, but it requires each node to exchange model updates with its neighbors in every round, incurring $\mathcal{O}(\text{degree} \times \text{model_size})$ communication per node per round. Therefore, existing approaches, whether centralized or decentralized, remain synchronization-heavy and costly for large foundation models. Furthermore, federated and gossip-based methods averaging the nodes' computation at each round are challenged by the bilinear structure of LoRA [7], [12], [14]. LoRA updates are of the form $W = BA$, see Section III. In federated and gossip-based methods, the factors A and B are aggregated separately, introducing a mismatch, since the product of averaged factors is generally not equal to the average of local updates.

To address these limitations, we advocate token-based random-walk learning as a scalable alternative. In this approach, a token representing the current model estimate moves randomly across the graph and is updated locally at each visited node [15]–[17]. Random-walk learning differs from consensus- and gossip-based methods in a fundamental way: it propagates a single mobile model rather than synchronizing many model copies. Each iteration requires only one node-to-node transfer, with communication on the order of the model size and independent of the network size. This substantially reduces communication and computational overhead, making the method well suited for large models. In addition, random-walk learning avoids this bilinear mismatch by eliminating factor-wise aggregation: the same LoRA token is sequentially updated and propagated across nodes, so the coupled structure of the low-rank factors is preserved throughout training.

Although random-walk learning has been studied extensively, its use for parameter-efficient fine-tuning remains largely unexplored. LoRA introduces a new challenge because its multiplicative low-rank structure, $W = W_0 + BA$ (see Section III-A), changes how model updates propagate along a random walk. In this paper, we propose a random-walk-based LoRA fine-tuning (RW-LoRA) scheme. At time t , the model token is located at node v_t , where the local fine-tuning

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objective is used to update the LoRA factors A and B . The updated token is then passed to the next node according to the random-walk transition rule. Our contributions are summarized as follows:

- (i) We propose, to the best of our knowledge, the first random-walk-based LoRA fine-tuning algorithm. We also provide rigorous convergence guarantees for non-convex objectives. This differs from many existing federated LoRA methods [7]–[12], which either lack convergence analysis or rely on stronger assumptions such as strong convexity.
- (ii) We conduct experiments on various NLP tasks and graph topologies, showing that the proposed method achieves strong task performance. **Moreover, RW-LoRA outperforms gossip-based LoRA because it avoids factor-wise aggregation and thereby prevents the bilinear mismatch. It also requires substantially less communication and computation to reach comparable accuracy.**

II. RELATED WORK

A. Federated and Decentralized Fine-Tuning of Foundation Models

LoRA has become a standard approach for adapting foundation models [6]. To enable efficient fine-tuning across distributed data sources, LoRA has recently been incorporated into federated learning. Representative methods include FLoRA [7], FedALT [8], FedLoRA [9], FRLoRA [10], FedSA-LoRA [11], FedEx-LoRA [12], and FedMomentum [13]. These methods mainly differ in how they aggregate or personalize LoRA adapters across clients. FLoRA supports heterogeneous LoRA adapters through a stacking-based aggregation scheme [7]. FedALT introduces a personalized federated LoRA framework in which each client maintains its own LoRA adapter while incorporating shared knowledge through a separate rest-of-world component [8]. FedLoRA addresses model heterogeneity by adding a homogeneous small adapter to each client’s heterogeneous local model and enabling global-local knowledge exchange [9]. FRLoRA performs global updates by combining the global model weights with a residual low-rank update [10]. FedSA-LoRA separates the roles of the two LoRA factors, sharing only the A matrices for aggregation while keeping the B matrices client-specific [11]. FedEx-LoRA introduces a residual error term to preserve exactness while maintaining LoRA’s computational and communication efficiency [12]. FedMomentum improves cross-round accumulation of LoRA updates through a structured, momentum-preserving aggregation scheme based on singular value decomposition [13]. Most existing federated LoRA methods still rely on centralized aggregation through a server. As a result, they inherit the communication and coordination bottlenecks of classical federated learning, which become increasingly limiting for large foundation models.

More recently, [14] studied fully decentralized and gossip-based LoRA fine-tuning algorithm, removing the need for a central server. They established convergence guarantees for non-convex objectives under standard assumptions.

In contrast, our work studies a random-walk-based LoRA method that avoids repeated server aggregation and neighbor-wise synchronization. This design reduces both communication and computational overhead.

B. Token-based Random-Walk Learning

Different from consensus- and gossip-based methods, token-based random-walk learning offers a different approach: a variable or model token moves across the graph according to a random walk and is updated using the local objective at each visited node. In this sense, each communication step directly moves the model to the next update location, rather than synchronizing several local models [15].

The literature on random-walk-based learning is broad, and we provide only a brief overview here. Seminal work in this area includes [16], [18], [19], which laid the foundation for optimization methods driven by Markov chain sampling. There are two directions to design and analyze token algorithms. First, [18] studied incremental optimization and subdifferentials methods under Markov chain sampling. Second, [16], [19] developed improved convergence guarantees for stochastic gradient and mirror-descent methods under Markovian sampling. A common feature of these results is that the convergence rate depends on the mixing time τ_{mix} of the underlying Markov chain, often through terms such as $\mathcal{O}(\frac{\tau_{\text{mix}}}{T}) + \mathcal{O}(\sqrt{\frac{\tau_{\text{mix}}}{T}})$, where T denotes the number of iterations. For a more complete account of random-walk-based learning and token algorithms, we refer the reader to [15].

III. PRELIMINARIES

A. Low-Rank Decomposition

Consider a pre-trained model Φ_0 and let $W_0 \in \mathbb{R}^{d_1 \times d_2}$ denote the weight matrix of one layer. In full fine-tuning, one would learn an update matrix $\Delta W \in \mathbb{R}^{d_1 \times d_2}$ and use $W = W_0 + \Delta W$. LoRA keeps W_0 fixed and restricts the update to a low-rank form [6] $\Delta W = BA$, where $A \in \mathbb{R}^{r \times d_2}$, $B \in \mathbb{R}^{d_1 \times r}$, and $r \ll \min\{d_1, d_2\}$. Thus, the adapted weight matrix becomes

$$W = W_0 + BA. \quad (1)$$

This reduces the number of trainable parameters from $d_1 d_2$ to $r(d_1 + d_2)$. Throughout the paper, we refer to r as the LoRA rank, which is typically chosen from $\{2, 4, 8, 16\}$. During the fine-tuning phase, LoRA optimizes factors A and B instead of directly updating ΔW , thus achieving substantial savings in GPU memory usage [7].

B. Random Walk

We now introduce the random walk model used throughout the paper.

Definition 1 (Communication topologies and RWs, Definition 3 in [20]). A communication topology is defined as a finite directed graph $\mathcal{G} \triangleq (\mathcal{V}, E)$ with the set of nodes $\mathcal{V} = [N]$ and the set of edges $E \subseteq \binom{\mathcal{V}}{2}$. A RW $\{v_t\}_{t \geq 0} : \Omega \rightarrow \mathcal{V}^+$ on this graph can be defined by the common transition probability

matrix $P : \mathcal{V} \rightarrow \mathcal{M}(\mathcal{V})$, where the probability of transition from node u to node v in one time step at time $t \in \mathbb{Z}_+$, is $P_{uv} \triangleq \Pr(v_{t+1} = v | v_t = u)$. We will call the RW on graph $\mathcal{G}$ aperiodic if transition probability matrix P is aperiodic.

Remark 1. Without loss of generality, we assume that $P_{uv} > 0$ for all nodes v connected to node u in graph $\mathcal{G}$. Therefore, transition matrix P is irreducible iff graph $\mathcal{G}$ is connected.

The stationary distribution describes the long-run fraction of time that the random walk spends at each node.

Definition 2 (Stationary distribution). Let P be the transition matrix defined in Definition 1. A probability π is called a stationary distribution if $\pi = \pi P$.

Let π_0 denote the initial distribution, and let $\pi_t = P^t \pi_0$ denote the distribution at time step t .

Definition 3 (Mixing time, Definition 2 in [15]). For $\epsilon > 0$, the mixing time $\tau_{\text{mix}}(\epsilon)$ of $\{X_t\}_{t \geq 0}$ is defined as:

$$\tau_{\text{mix}}(\epsilon) = \inf \{t \geq 1 | \forall \pi_0, d_{\text{TV}}(P^t \pi_0, \pi) \leq \epsilon\},$$

where d_{TV} is the total-variation distance and π_0 is the initial distribution.

C. Random Walk Learning

We consider the decentralized stochastic optimization

$$\min_{\mathbf{w}} f(\mathbf{w}) = \min_{\mathbf{w}} \mathbb{E}_{v \sim \pi} [f_v(\mathbf{w})], \quad \mathbf{w} \in \mathbb{R}^d, \quad (2)$$

where π is a probability distribution over the node set $\mathcal{V}$, and f_v denotes the local objective at node v . In our setting, f is the global fine-tuning objective, while f_v is the local fine-tuning objective associated with the data stored at node v .

Let $\{v_t\}_{t \geq 0}$ be a random walk on $\mathcal{V}$ with stationary distribution π . At time t , the model token is located at node v_t and is updated using only the local objective at that node [15]–[17]:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f_{v_t}(\mathbf{w}_t), \quad (3)$$

where $\eta > 0$ is the stepsize.

IV. RANDOM-WALK-BASED LORA FINE-TUNING SCHEME

Let $\mathcal{G}$ be defined in Definition 1. The network collaboratively aims to solve the following optimization problem:

$$\min_{A, B} f(W) = \min_{A, B} \mathbb{E}_{v \sim \pi} [f_v(W)], \quad (4)$$

where $f_v : \mathbb{R}^{d_1 \times d_2} \rightarrow \mathbb{R}$ is the local fine-tuning objective of node v , and π is a pre-determined target distribution. Let $\mathcal{D}_v$ denote the local data distribution specific to user v , then the local fine-tuning objective f_v is given in stochastic form:

$$f_v(W) = \mathbb{E}_{\xi_v \sim \mathcal{D}_v} [F_v(W; \xi_v)]. \quad (5)$$

For simplicity, we use ∇ to denote the gradient with respect to W , i.e., ∇_W , and use ∇_A and ∇_B to denote the gradients with respect to A and B , respectively. We define

$$\nabla \tilde{f}_v(W) \triangleq \nabla F_v(W; \xi_v). \quad (6)$$

Algorithm 1 RW-LoRA Algorithm

Require: Graph $\mathcal{G} = (\mathcal{V}, E)$, transition matrix P , stepsize η , LoRA rank r , number of iterations T , pre-trained weight matrix $W_0 \in \mathbb{R}^{d_1 \times d_2}$

- 1: Initialize $A^{(0)} \sim \mathcal{N}(0, \sigma^2)$ and $B^{(0)} = 0$
- 2: Initialize the random walk at node $v_0 \in \mathcal{V}$
- 3: **for** $t = 0, 1, \dots, T - 1$, **at node** v_t **do**
- 4: Form the LoRA-adapted weight $W^{(t)} = W_0 + B^{(t)} A^{(t)}$.
- 5: Update the LoRA matrices via (7) and (8).
- 6: Sample the next node according to $v_{t+1} \sim P_{v_t, \cdot}$.
- 7: Transfer the token $(A^{(t+1)}, B^{(t+1)})$ to node v_{t+1} .
- 8: **end for**
- 9: **return** $A^{(T)}, B^{(T)}$ and $W^{(T)} = W_0 + B^{(T)} A^{(T)}$

Given a target sampling distribution π (assuming $\pi_v > 0$ for all $v \in \mathcal{V}$), the Metropolis-Hastings algorithm [16] provides a principled way to construct a RW on $\mathcal{G}$ with a transition matrix P such that π is the stationary distribution of the Markov chain defined by P .¹ Let $\{v_t\}_{t \geq 0}$ be a random walk on $\mathcal{V}$ that evolves according to the transition matrix P . Although (3) is written for vector-valued parameters, LoRA fine-tuning involves matrix-valued parameters. We therefore apply a similar random-walk updating rule to the two LoRA matrices A and B : At time t , the model token is located at node v_t , and the local fine-tuning objective at that node is used to update the LoRA matrices,

$$A^{(t+1)} = A^{(t)} - \eta \nabla_A \tilde{f}_{v_t}(W^{(t)}), \quad (7)$$

$$B^{(t+1)} = B^{(t)} - \eta \nabla_B \tilde{f}_{v_t}(W^{(t)}), \quad (8)$$

where the initial weights $A^{(0)} \sim \mathcal{N}(0, \sigma^2)$, $B^{(0)} = 0$, and $W^{(0)} = B^{(0)} A^{(0)}$, as adopted in [4]. The algorithm is outlined in Algorithm 1.

V. THEORETICAL RESULTS

A. Optimization Metric

Most existing convergence analyses for distributed LoRA fine-tuning, such as [8], [9], [11], measure stationarity through $\nabla f(W^{(t)})$. This treatment is convenient, but it does not fully reflect the optimization structure of LoRA. In LoRA fine-tuning, the trainable variables are not the full update matrix ΔW directly, but its low-rank factors A and B , where $\Delta W = BA$. Therefore, the algorithm updates A and B separately, and the resulting change in ΔW is determined by their coupled product.

This distinction is important for convergence analysis. A small gradient with respect to the full parameter $W^{(t)}$ does not directly characterize stationarity in the actual optimization variables A and B [4]. Conversely, because of the factorized parameterization, the relationship between the gradients in (A, B) and the gradient with respect to $W^{(t)}$ is nontrivial [21].

¹The transition matrix P depends on both the target sampling distribution π and the graph $\mathcal{G}$. We re-parameterize it as $P_{\pi, \mathcal{G}} \in \mathbb{R}_+^{N \times N}$, and simply write P when there is no risk of confusion.

Thus, for a LoRA algorithm whose updates are performed on A and B , stationarity should be measured with respect to these factors rather than only through $\nabla f(W^{(t)})$.

Following [4], we therefore use the stationarity measure

$$\mathbb{E} \left[\|\nabla_A f(W^{(t)})\| \right] + \mathbb{E} \left[\|\nabla_B f(W^{(t)})\| \right], \quad (9)$$

which directly evaluates whether the actual trainable LoRA factors are approaching a stationary point.

Through a detailed convergence analysis, we show that the proposed RW-LoRA algorithm converges to a stationary point with respect to the measure defined in (9). The complete proofs are provided in Appendix A. We begin by introducing several additional assumptions.

B. Convergence Analysis

Assumption 1 (Submultiplicative property). *The matrix norm $\|\cdot\|$ used in this paper is submultiplicative, i.e., $\|BA\| \leq \|B\|\|A\|$ for any compatible matrices A and B .*

This assumption is mild and is satisfied by most commonly used matrix norms, such as the spectral norm, Frobenius norm, 1-norm, and ∞ -norm.

Assumption 2 (L -smooth, Assumption 4.1 in [4]). *We assume that each local fine-tuning objective f_v is L -smooth, i.e., for all $W, W' \in \mathbb{R}^{d_1 \times d_2}$, we have*

$$\|\nabla f_v(W) - \nabla f_v(W')\| \leq L\|W - W'\|.$$

Assumption 3 (Bounded gradient, Assumption 4.2 in [4]). *We assume that the stochastic gradients are unbiased and that their expected squared norm remains uniformly bounded:*

$$\begin{aligned} \mathbb{E}_{\xi_v \sim \mathcal{D}_v} [\nabla \tilde{f}_v(W)] &= \nabla f_v(W), \\ \mathbb{E}_{\xi_v \sim \mathcal{D}_v} [\|\nabla \tilde{f}_v(W)\|^2] &\leq c^2, \quad v \in \mathcal{V}, \end{aligned}$$

where ξ_v represents a randomly sampled subset of training data from v -th node.

Assumption 3 is standard in decentralized optimization [4], [7], [9], [11]. It helps bound the magnitude of the update steps, as in the analysis in Lemmas 7 and 11 of [4]. For notational simplicity, we omit the subscript $\xi_v \sim \mathcal{D}_v$ when it is clear from the context.

Assumption 4 (Bounded variance, Assumption 5.1 in [22]). *There exists σ^2 such that for all $v \in \mathcal{V}$ and all $W \in \mathbb{R}^{d_1 \times d_2}$, we have:*

$$\mathbb{E} [\|\nabla \tilde{f}_v(W) - \nabla f(W)\|^2] \leq \sigma^2.$$

In Assumption 4, the expectation $\mathbb{E}$ is taken over two sources of randomness: (i) the randomness in data sampling at node v , i.e., $\xi_v \sim \mathcal{D}_v$, and (ii) the randomness in the global objective, as defined in (4). It is used to control the stochastic noise arising from the random-walk sampling process [22, Lemmas C.1 & C.2].

Assumptions 3 and 4 are both necessary and play distinct roles in our analysis. For vector-valued optimization variables,

one of these assumptions is often sufficient, since the other is not needed for the convergence proof. In our setting, however, the optimization variables are the matrix-valued LoRA factors A and B , with $\Delta W = BA$. Because A and B are updated separately, dropping either assumption would prevent us from completing the convergence proof.

Assumption 5 (Assumption 4.3 in [4]). *There exists constants $a > 0$ and $b > 0$ such that: $\|A^{(t)}\| \leq a$, $\|B^{(t)}\| \leq b$ for all $0 \leq t \leq T$.*

In Assumption 5, the constants a and b are chosen uniformly over t . These constants may depend on the matrix dimensions, namely r , d_1 , and d_2 .

Theorem 1. *Let Assumptions 1 ~ 5 hold. Let $\epsilon < \frac{1}{2}$, $\pi_{\min} = \min_{v \in \mathcal{V}} \{\pi_v\}$, $\tau \geq \tau_{\text{mix}}(\pi_{\min}\epsilon)$, $T \geq \tau^2$, $\eta \leq \frac{1}{4L\sqrt{T}}$, we obtain²:*

$$\begin{aligned} &\frac{1}{T} \sum_{t=1}^T \left(\mathbb{E} [\|\nabla_A f(W^{(t)})\|^2] + \mathbb{E} [\|\nabla_B f(W^{(t)})\|^2] \right) \\ &\leq \tilde{\mathcal{O}} \left(\epsilon^2 + \frac{L\tau}{\sqrt{T}} + \frac{L\tau}{T} \right). \end{aligned} \quad (10)$$

Theorem 1 shows that the convergence rate depends on the mixing time τ of the underlying Markov chain. In particular, our bound has the form $\tilde{\mathcal{O}} \left(\epsilon^2 + \frac{L\tau}{\sqrt{T}} + \frac{L\tau}{T} \right)$. This is broadly consistent with the rates obtained in standard random-walk learning for vector-valued objectives, where the dependence on the number of iterations typically appears as $\mathcal{O} \left(\frac{\tau_{\text{mix}}}{T} \right) + \mathcal{O} \left(\sqrt{\frac{\tau_{\text{mix}}}{T}} \right)$ [16], [18], [19]. There are, however, two differences. First, our result is stated in $\tilde{\mathcal{O}}$ notation, whereas the standard vector-valued setting is often expressed in $\mathcal{O}$ notation. Second, the coefficient of the $1/\sqrt{T}$ term scales linearly with the mixing time τ in our LoRA setting, while it scales as $\sqrt{\tau_{\text{mix}}}$ in the standard vector-valued case. This difference arises from the low-rank matrix factorization in LoRA: the trainable variables are the matrices A and B , and the update $\Delta W = BA$ introduces additional coupling between the two factors.

VI. EXPERIMENTS

A. Experiment Settings

1) **Datasets:** We evaluate the proposed RW-LoRA algorithm on five GLUE datasets, covering two types of tasks:

- (i) **Sentence-pair classification:** We fine-tune the base model on MRPC [23], QQP [24], QNLI [24], [25], and MNLI [24], [26]. These tasks require the model to determine semantic equivalence, textual entailment, or whether a sentence contains the answer to a given question.
- (ii) **Sentiment classification:** We fine-tune the base model on SST-2 [27], where the task is to classify each sentence as positive or negative.

² $\tilde{\mathcal{O}}$ hides logarithmic factors.

2) *Baseline*: Decentralized federated learning for LoRA remains largely unexplored. To the best of our knowledge, [14] is the closest existing work to our setting. We therefore use its gossip-based federated LoRA method as our baseline.

3) *Setup*: We use RoBERTa-base [28] as the backbone model for all tasks. Following [14], we set the LoRA rank to $r = 16$. We use the default ADAMW optimizer with learning rate 10^{-3} . The random-walk fine-tuning process is run over 10 nodes. At each communication round, the active node performs 1 local training steps with batch size $b = 32$ before passing the model token to the next node. We run 325 rounds for all NLP tasks.

4) *Topologies*: As previously discussed, in our decentralized framework, nodes communicate exclusively along the edges of a fixed communication graph that connects 10 nodes. We considered two widely used network topologies in communication and edge computing applications [20], [29]: (i) ring graph and (ii) complete graph.

B. Experimental Results

a) *Task Performance*.: The task performance of random-walk-based LoRA is reported in Tables I & II. Table I reports results on the complete graph, while Table II reports results on the ring topology³. In both tables, *Final Score* denotes the model performance at the last training round, whereas *Best Score* denotes the highest performance achieved over all training rounds.

	MRPC	SST2	QNLI	MNLI	QQP
RW (Final Scores)	89.98	94.15	91.09	84.66	88.84
Gossip (Final Scores)	83.57	93.23	90.17	83.66	86.71
RW (Best Scores)	91.49	94.50	92.02	86.20	88.92
Gossip (Best Scores)	83.57	93.23	90.17	84.06	86.71

TABLE I: Accuracy under Complete Graph

	MRPC	SST2	QNLI	MNLI	QQP
RW (Final Scores)	90.75	92.43	90.96	85.18	88.54
Gossip (Final Scores)	88.19	92.89	91.23	85.76	88.48
RW (Best Scores)	93.02	94.50	92.00	86.07	88.93
Gossip (Best Scores)	89.30	93.58	91.76	85.76	88.58

TABLE II: Accuracy under Ring Graph

On the complete graph, the proposed RW-LoRA algorithm outperforms gossip-based LoRA algorithm in both final and best accuracy. On the ring topology, the RW-LoRA algorithm achieves comparable final accuracy and higher best accuracy than gossip-based LoRA algorithm. Overall, these results show that the RW-LoRA algorithm improves upon the task performance of gossip-based LoRA algorithm. **This improvement comes from RW-LoRA avoiding factor-wise aggregation, which prevents the bilinear mismatch observed in gossip-based LoRA.**

b) *Communication and Computation Efficiency*.: Under gossip-based LoRA, each node exchanges updates with its

³In both tables, the performance metric for MRPC is F1, whereas the metric for all other datasets is accuracy. For simplicity of presentation, we use the term ‘‘Accuracy’’ in the tables to refer to the corresponding task performance metric.

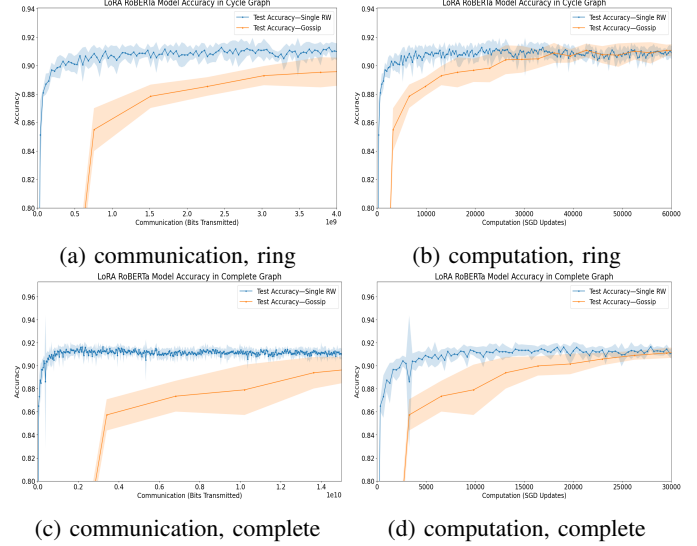


Fig. 1: Accuracy v.s communication and computation overhead in different graphs.

neighbors in every communication round. Thus, the per-node communication cost is $O(\text{degree} \times \text{model_size})$, and the total network-wide cost per round is $O(\text{network_size} \times \text{degree} \times \text{model_size})$. In contrast, random walk-based LoRA involves only one active transmission per round, from the current node to the next node on the walk. Its communication cost per round is therefore $O(\text{model_size})$. Since each active transmission is accompanied by a local update, the computation cost scales with the communication cost.

Fig. 1 show the results on the QNLI dataset. We evaluate all methods on five datasets, and the accuracy curves exhibit similar trends across them. Due to space constraints, we report the accuracy curves for QNLI only. In Figs. 1a and 1c, the x -axis denotes the communication overhead, while in Figs. 1b and 1d, the x -axis denotes the computation overhead. The results show that the RW-LoRA algorithm reaches comparable or better accuracy with substantially lower communication and computation cost than the gossip-based LoRA algorithm. This suggests that the RW-LoRA algorithm is more resource-efficient, especially when communication and local computation are the main bottlenecks.

VII. CONCLUSION

In this paper, we studied decentralized a random-walk-based LoRA framework. Unlike existing federated and gossip-based approaches that rely on multiple synchronized model copies, our method maintains a single mobile model that is updated sequentially across the network. We established convergence guarantees for non-convex objectives. Experimental results demonstrate that the proposed method achieves competitive performance while being substantially more resource-efficient than the gossip-based LoRA algorithm.

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APPENDIX A
PROOF OF THEOREM 1

The proof consists of three parts. In the first part, we establish several useful lemmas. In the second part, we decompose the critical term $f(W^{(t+1)}) = f(B^{(t+1)}A^{(t+1)})$. In the third part, we apply a telescoping argument to obtain the desired result.

A. Useful lemmas

First of all, we provide the following useful lemmas.

Lemma 1. *The expected squared norms of $\nabla_A \tilde{f}_v(W)$ and $\nabla_B \tilde{f}_v(W)$ are bounded:*

$$\mathbb{E} \left[\|\nabla_A \tilde{f}_v(W)\|^2 \right] \leq b^2 c^2, \quad (11)$$

$$\mathbb{E} \left[\|\nabla_B \tilde{f}_v(W)\|^2 \right] \leq a^2 c^2. \quad (12)$$

Proof. Note that $W = BA$, according to (6), we have:

$$\begin{aligned} \nabla_A \tilde{f}_v(W) &= B^T \nabla \tilde{f}_v(W), \\ \nabla_B \tilde{f}_v(W) &= \nabla \tilde{f}_v(W) A^T. \end{aligned} \quad (13)$$

Taking norms on both sides of (13), and using the submultiplicative property in Assumption 1 together with Assumption 5, we obtain

$$\|\nabla_A \tilde{f}_v(W)\| \leq \|B\| \|\nabla \tilde{f}_v(W)\| \leq b \|\nabla \tilde{f}_v(W)\|, \quad (14)$$

$$\|\nabla_B \tilde{f}_v(W)\| \leq \|A\| \|\nabla \tilde{f}_v(W)\| \leq a \|\nabla \tilde{f}_v(W)\|. \quad (15)$$

According to Assumption 3, taking square and expectations on both sides of (14) and (15), we obtain:

$$\mathbb{E} \left[\|\nabla_A \tilde{f}_v(W)\|^2 \right] \leq b^2 c^2,$$

$$\mathbb{E} \left[\|\nabla_B \tilde{f}_v(W)\|^2 \right] \leq a^2 c^2.$$

□

Lemma 2. *The expected squared differences $\mathbb{E}[\|A^{(t+1)} - A^{(t)}\|^2]$, $\mathbb{E}[\|B^{(t+1)} - B^{(t)}\|^2]$, and $\mathbb{E}[\|W^{(t+1)} - W^{(t)}\|^2]$ are upper bounded as follows:*

$$\mathbb{E} \left[\|A^{(t+1)} - A^{(t)}\|^2 \right] \leq \eta^2 b^2 c^2, \quad (16)$$

$$\mathbb{E} \left[\|B^{(t+1)} - B^{(t)}\|^2 \right] \leq \eta^2 a^2 c^2, \quad (17)$$

$$\mathbb{E} \left[\|W^{(t+1)} - W^{(t)}\|^2 \right] \leq 3(\eta^2 b^4 c^2 + \eta^2 a^4 c^2 + \eta^4 a^2 b^2 c^4). \quad (18)$$

Proof. According to (7) and (8), we have:

$$\begin{aligned} \|A^{(t+1)} - A^{(t)}\|^2 &= \eta^2 \|\nabla_A \tilde{f}_{v_t}(W^{(t)})\|^2, \\ \|B^{(t+1)} - B^{(t)}\|^2 &= \eta^2 \|\nabla_B \tilde{f}_{v_t}(W^{(t)})\|^2. \end{aligned}$$

Taking expectations on both sides of above equalities, and based on (11) and (12), we have (16) and (17).

Next, we calculate $W^{(t+1)} - W^{(t)}$:

$$\begin{aligned} W^{(t+1)} - W^{(t)} &= B^{(t+1)} A^{(t+1)} - B^{(t)} A^{(t)} \\ &= \left(A^{(t)} - \eta \nabla_A \tilde{f}_{v_t}(W^{(t)}) \right) \left(B^{(t)} - \eta \nabla_B \tilde{f}_{v_t}(W^{(t)}) \right) - B^{(t)} A^{(t)} \\ &= -\eta \nabla_A \tilde{f}_{v_t}(W^{(t)}) B^{(t)} - \eta A^{(t)} \nabla_B \tilde{f}_{v_t}(W^{(t)}) + \eta^2 \nabla_A \tilde{f}_{v_t}(W^{(t)}) \nabla_B \tilde{f}_{v_t}(W^{(t)}). \end{aligned}$$

Taking norms on both sides, by the triangle inequality, we obtain:

$$\begin{aligned} \|W^{(t+1)} - W^{(t)}\| &\leq \eta \|\nabla_A \tilde{f}_{v_t}(W^{(t)}) B^{(t)}\| + \eta \|A^{(t)} \nabla_B \tilde{f}_{v_t}(W^{(t)})\| \\ &\quad + \eta^2 \|\nabla_A \tilde{f}_{v_t}(W^{(t)}) \nabla_B \tilde{f}_{v_t}(W^{(t)})\|. \end{aligned}$$

Applying the submultiplicative property, we obtain

$$\begin{aligned}\|W^{(t+1)} - W^{(t)}\| &\leq \eta b \|\nabla_A \tilde{f}_{v_t}(W^{(t)})\| + \eta a \|\nabla_B \tilde{f}_{v_t}(W^{(t)})\| \\ &\quad + \eta^2 \|\nabla_A \tilde{f}_{v_t}(W^{(t)})\| \|\nabla_B \tilde{f}_{v_t}(W^{(t)})\|.\end{aligned}$$

By Cauchy–Schwarz inequality, it follows that

$$\begin{aligned}\|W^{(t+1)} - W^{(t)}\|^2 &\leq 3\eta^2 b^2 \|\nabla_A \tilde{f}_{v_t}(W^{(t)})\|^2 + 3\eta^2 a^2 \|\nabla_B \tilde{f}_{v_t}(W^{(t)})\|^2 \\ &\quad + 3\eta^4 \|\nabla_A \tilde{f}_{v_t}(W^{(t)})\|^2 \|\nabla_B \tilde{f}_{v_t}(W^{(t)})\|^2.\end{aligned}$$

Taking expectations on both sides of above equalities, based on (16) and (17), we obtain:

$$\mathbb{E} \left[\|W^{(t+1)} - W^{(t)}\|^2 \right] \leq 3(\eta^2 b^4 c^2 + \eta^2 a^4 c^2 + \eta^4 a^2 b^2 c^4).$$

□

Lemma 3 (Lemma 4.10 in [4]). *Under Assumption 2, each local fine-tuning objective f_v is Lb^2 -smooth with respect to A when B is fixed, and La^2 -smooth with respect to B when A is fixed.*

Lemma 4. *Let $f(\cdot)$ be defined in (4). Then, f is L -smooth.*

Proof. According to (4),

$$\nabla f(W) - \nabla f(W') = \mathbb{E}_{v \sim \pi} [\nabla f_v(W) - \nabla f_v(W')].$$

By the triangle inequality, we have:

$$\|\nabla f(W) - \nabla f(W')\| = \|\mathbb{E}_{v \sim \pi} [\nabla f_v(W) - \nabla f_v(W')]\| \leq \mathbb{E}_{v \sim \pi} \|\nabla f_v(W) - \nabla f_v(W')\|.$$

Based on Assumption 2, it follows that

$$\|\nabla f(W) - \nabla f(W')\|_F \leq \mathbb{E}_{v \sim \pi} [L\|W - W'\|] = L\|W - W'\|.$$

□

Lemma 5. *Let $f(\cdot)$ be defined in (4). Then, f is Lb^2 -smooth with respect to A when B is fixed, and La^2 -smooth with respect to B when A is fixed.*

Proof. The proof directly follows from Lemma 3 and Lemma 4. □

Let π_0 denote the initial distribution, and let $\pi_t = P^t \pi_0$ denote the distribution at time step t . Let $\pi_{\min} = \min_{u \in \mathcal{V}} \pi_u$.

Lemma 6. *For $t \geq 0$ and if $v_t \sim \pi_t$ for $d_{TV}(\pi_t, \pi) \leq \frac{\pi_{\min}}{2}$, we have*

$$\mathbb{E} \left[\|\nabla \tilde{f}_{v_t}(W)\|^2 \right] \leq 3\sigma^2 + 2\mathbb{E} [\|\nabla f(W)\|^2]$$

Proof. Since $d_{TV}(\pi_t, \pi) \leq \frac{\pi_{\min}}{2}$, then for any $v \in \mathcal{V}$, we have

$$\Pr(v_t = v) \leq \pi_v + \pi_v/2 = \frac{3\pi_v}{2}.$$

By the triangle inequality, Cauchy–Schwarz inequality, and Assumption 4, it follows that

$$\begin{aligned}\mathbb{E} \left[\|\nabla \tilde{f}_{v_t}(W)\|^2 \right] &\leq 2\mathbb{E} \left[\|\nabla \tilde{f}_{v_t}(W) - \nabla f(W)\|^2 \right] + 2\mathbb{E} [\|\nabla f(W)\|^2] \\ &\leq 2 \sum_{v \in \mathcal{V}} \Pr(v_t = v) \sigma^2 + 2\mathbb{E} [\|\nabla f(W)\|^2] \\ &= 3\sigma^2 + 2\mathbb{E} [\|\nabla f(W)\|^2].\end{aligned}$$

□

Lemma 7. *The expected squared norms of $\nabla f(W)$, $\nabla_A f(W)$ and $\nabla_B f(W)$ are bounded:*

$$\mathbb{E} [\|\nabla f(W)\|^2] \leq 2\sigma^2 + 2c^2, \tag{19}$$

$$\mathbb{E} [\|\nabla_A f(W)\|^2] \leq 2b^2(\sigma^2 + c^2), \tag{20}$$

$$\mathbb{E} [\|\nabla_B f(W)\|^2] \leq 2a^2(\sigma^2 + c^2). \tag{21}$$

Proof. By the triangle inequality and Cauchy–Schwarz inequality, we have

$$\mathbb{E} [\|\nabla f(W)\|^2] \leq 2\mathbb{E} [\|\nabla \tilde{f}_v(W) - \nabla f(W)\|^2] + 2\mathbb{E} [\|\nabla \tilde{f}_v(W)\|^2]$$

According to Assumption 3 and Assumption 4, we have

$$\mathbb{E} [\|\nabla f(W)\|^2] \leq 2\sigma^2 + 2c^2.$$

Note that $W = BA$, we have:

$$\nabla_A f(W) = B^T \nabla f(W), \quad \nabla_B f(W) = \nabla f(W) A^T. \quad (22)$$

Applying the submultiplicative property and taking expectations on (22), we obtain:

$$\begin{aligned} \mathbb{E} [\|\nabla_A f(W)\|^2] &\leq 2b^2(\sigma^2 + c^2) \\ \mathbb{E} [\|\nabla_B f(W)\|^2] &\leq 2a^2(\sigma^2 + c^2). \end{aligned}$$

B. Decomposition of $\mathbb{E} [f(B^{(t+1)} A^{(t+1)})]$

Since f is smooth, by the Descent Lemma, we have:

$$\begin{aligned} \mathbb{E} [f(B^{(t+1)} A^{(t+1)})] &\leq \mathbb{E} [f(B^{(t+1)} A^{(t)})] + \mathbb{E} \langle \nabla_A f(B^{(t+1)} A^{(t)}), A^{(t+1)} - A^{(t)} \rangle \\ &\quad + \frac{Lb^2}{2} \mathbb{E} [\|A^{(t+1)} - A^{(t)}\|^2]. \end{aligned} \quad (23)$$

In (23), the expectation $\mathbb{E}$ is taken over the target sampling π and the randomness from the position of the RW at time t , i.e., v_t . Again, by the Descent Lemma, we further have:

$$\begin{aligned} \mathbb{E} [f(B^{(t+1)} A^{(t)})] &\leq \mathbb{E} [f(B^{(t)} A^{(t)})] + \mathbb{E} \langle \nabla_B f(B^{(t)} A^{(t)}), B^{(t+1)} - B^{(t)} \rangle \\ &\quad + \frac{La^2}{2} \mathbb{E} [\|B^{(t+1)} - B^{(t)}\|^2]. \end{aligned} \quad (24)$$

Combine (23) and (24), we obtain:

$$\begin{aligned} \mathbb{E} [f(B^{(t+1)} A^{(t+1)})] &\leq \mathbb{E} [f(B^{(t)} A^{(t)})] + \mathbb{E} \langle \nabla_A f(B^{(t+1)} A^{(t)}), A^{(t+1)} - A^{(t)} \rangle \\ &\quad + \mathbb{E} \langle \nabla_B f(B^{(t)} A^{(t)}), B^{(t+1)} - B^{(t)} \rangle + \frac{Lb^2}{2} \mathbb{E} [\|A^{(t+1)} - A^{(t)}\|^2] \\ &\quad + \frac{La^2}{2} \mathbb{E} [\|B^{(t+1)} - B^{(t)}\|^2]. \end{aligned} \quad (25)$$

By Cauchy–Schwarz inequality, we have

$$\begin{aligned} \mathbb{E} \langle \nabla_A f(B^{(t+1)} A^{(t)}), A^{(t+1)} - A^{(t)} \rangle &\leq \frac{1}{2} \mathbb{E} [\|\nabla_A f(B^{(t+1)} A^{(t)})\|^2] \\ &\quad + \frac{1}{2} \mathbb{E} [\|A^{(t+1)} - A^{(t)}\|^2]. \end{aligned}$$

From (16) in Lemma 2 and (20) in Lemma 7,

$$\mathbb{E} \langle \nabla_A f(B^{(t+1)} A^{(t)}), A^{(t+1)} - A^{(t)} \rangle \leq b^2(\sigma^2 + c^2) + \eta^2 \frac{b^2 c^2}{2}. \quad (26)$$

According to the update rule (8), we have

$$\begin{aligned} \mathbb{E} \langle \nabla_B f(B^{(t)} A^{(t)}), B^{(t+1)} - B^{(t)} \rangle &= \mathbb{E} \langle \nabla_B f(B^{(t)} A^{(t)}), -\eta \nabla_B \tilde{f}_{v_t}(W^{(t)}) \rangle \\ &= -\eta \mathbb{E} \langle \nabla_B f(B^{(t)} A^{(t)}), \nabla_B \tilde{f}_{v_t}(W^{(t)}) \rangle \\ &= \mathbb{E} \left[-\eta \langle \nabla_B f(W^{(t)}), \nabla_B \tilde{f}_{v_t}(W^{(t)}) \rangle \right]. \end{aligned} \quad (27)$$

Let $\epsilon < \frac{1}{2}$, $\tau \geq \tau_{\text{mix}}(\pi_{\min}\epsilon)$, and $t \geq \tau$, based on [22, Appendix C.1], we have:

$$\begin{aligned} \mathbb{E} \left[-\eta \left\langle \nabla_B \tilde{f}_{v_t}(W^{(t)}), \nabla_B f(W^{(t)}) \right\rangle \right] &= \mathbb{E} \left[-\eta \left\langle \nabla_B \tilde{f}_{v_t}(W^{(t-\tau)}), \nabla_B f(W^{(t-\tau)}) \right\rangle \right] \\ &\quad + \mathbb{E} \left[-\eta \left\langle \nabla_B \tilde{f}_{v_t}(W^{(t)}), \nabla_B f(W^{(t)}) - \nabla_B f(W^{(t-\tau)}) \right\rangle \right] \\ &\quad + \mathbb{E} \left[-\eta \left\langle \nabla_B \tilde{f}(W^{(t)}) - \nabla_B \tilde{f}(W^{(t-\tau)}), \nabla_B f(W^{(t-\tau)}) \right\rangle \right] \\ &\triangleq E_1 + E_2 + E_3. \end{aligned}$$

By the same analysis as in [22, Appendix C.1], we obtain the following three inequalities: (28), (29), and (30). First,

$$E_1 \leq -\frac{\eta}{4} \mathbb{E} \left[\|\nabla_B f(W^{(t-\tau)})\|^2 \right] + \eta \epsilon^2 \sigma^2. \quad (28)$$

By utilizing Assumption 2, we have:

$$E_2 \leq \frac{\eta^2 L}{2} \tau \mathbb{E} \left[\|\nabla_B \tilde{f}_{v_t}(W^{(t)})\|^2 \right] + \frac{\eta^2 L}{2} \sum_{s=t-\tau}^{t-1} \mathbb{E} \left[\|\nabla_B \tilde{f}_{v_s}(W^{(s)})\|^2 \right]. \quad (29)$$

Similarly, we have:

$$E_3 \leq \frac{\eta^2 L}{2} \tau \mathbb{E} \left[\|\nabla_B f(W^{(t-\tau)})\|^2 \right] + \frac{\eta^2 L}{2} \sum_{s=t-\tau}^{t-1} \mathbb{E} \left[\|\nabla_B \tilde{f}_{v_s}(W^{(s)})\|^2 \right] \quad (30)$$

Based on Lemma 2, Lemma 7, and (26) $\sim$ (30), we have:

$$\begin{aligned} \mathbb{E} \left[f(B^{(t+1)} A^{(t+1)}) \right] &\leq \mathbb{E} \left[f(B^{(t)} A^{(t)}) \right] + \eta \epsilon^2 \sigma^2 + b^2(\sigma^2 + c^2) \\ &\quad + \eta^2 \left(\frac{3L}{2} \tau a^2 c^2 + \frac{c^2 L}{2} (a^4 + b^4) + \frac{b^2 c^2}{2} \right) \\ &\quad - \left(\frac{\eta}{4} - \frac{\eta^2 L}{2} \tau \right) \mathbb{E} \left[\|\nabla_B f(W^{(t-\tau)})\|^2 \right]. \end{aligned} \quad (31)$$

In (23) and (24), we decompose $f(B^{(t+1)} A^{(t+1)})$ using the intermediate terms $f(B^{(t)} A^{(t+1)})$ and $f(B^{(t)} A^{(t)})$. By a similar analysis, we obtain:

$$\begin{aligned} \mathbb{E} \left[f(B^{(t+1)} A^{(t+1)}) \right] &\leq \mathbb{E} \left[f(B^{(t)} A^{(t)}) \right] + \eta \epsilon^2 \sigma^2 + a^2(\sigma^2 + c^2) \\ &\quad + \eta^2 \left(\frac{3L}{2} \tau b^2 c^2 + \frac{c^2 L}{2} (a^4 + b^4) + \frac{a^2 c^2}{2} \right) \\ &\quad - \left(\frac{\eta}{4} - \frac{\eta^2 L}{2} \tau \right) \mathbb{E} \left[\|\nabla_A f(W^{(t-\tau)})\|^2 \right]. \end{aligned} \quad (32)$$

Summing over (31) and (32), we derive:

$$\begin{aligned} \mathbb{E} \left[f(B^{(t+1)} A^{(t+1)}) \right] &\leq \mathbb{E} \left[f(B^{(t)} A^{(t)}) \right] + \eta \epsilon^2 \sigma^2 + \frac{a^2 + b^2}{2} (\sigma^2 + c^2) \\ &\quad + \eta^2 \left(\frac{3(a^2 + b^2)L}{4} \tau c^2 + \frac{c^2 L}{2} (a^4 + b^4) + \frac{c^2(a^2 + b^2)}{4} \right) \\ &\quad - \left(\frac{\eta}{4} - \frac{\eta^2 L}{2} \tau \right) \left(\mathbb{E} \left[\|\nabla_A f(W^{(t-\tau)})\|^2 \right] + \mathbb{E} \left[\|\nabla_B f(W^{(t-\tau)})\|^2 \right] \right). \end{aligned} \quad (33)$$

C. Telescoping

Let $T \geq \tau^2$, $\eta \leq \frac{1}{4L\sqrt{T}}$, we have

$$\frac{\eta}{8} \leq \frac{\eta}{4} - \frac{\eta^2 L}{2} \tau.$$

Then, summing for $\tau \leq t < T + \tau$, by telescoping (33), we obtain:

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \left(\mathbb{E} \left[\|\nabla_A f(W^{(t)})\|^2 \right] + \mathbb{E} \left[\|\nabla_B f(W^{(t)})\|^2 \right] \right) \\ \leq \epsilon^2 \sigma^2 + \eta \Gamma + \frac{8\mathbb{E} [f(W^{(\tau)})] + 4(a^2 + b^2)(\sigma^2 + c^2)}{\eta T}, \end{aligned} \quad (34)$$

where

$$\Gamma = 6(a^2 + b^2)L\tau c^2 + 4c^2L(a^4 + b^4) + 2c^2(a^2 + b^2).$$

Now, we upper bound $\mathbb{E} [f(W^{(\tau)})]$. According to Lemma 4, f is L -smooth. By the Descent Lemma, we have:

$$f(W^{(\tau)}) - f(W^{(\tau-1)}) \leq \langle \nabla f(W^{(\tau-1)}), W^{(\tau)} - W^{(\tau-1)} \rangle + \frac{L}{2} \|W^{(\tau)} - W^{(\tau-1)}\|^2.$$

By Cauchy–Schwarz inequality,

$$f(W^{(\tau)}) - f(W^{(\tau-1)}) \leq \left(\|\nabla f(W^{(\tau-1)})\| + \frac{L}{2} \right) \|W^{(\tau)} - W^{(\tau-1)}\|^2,$$

which implies

$$\mathbb{E} [f(W^{(\tau)})] - \mathbb{E} [f(W^{(\tau-1)})] \leq \left(\sqrt{\mathbb{E} [\|\nabla f(W^{(\tau-1)})\|^2]} + \frac{L}{2} \right) \|W^{(\tau)} - W^{(\tau-1)}\|^2.$$

Based on (19) in Lemma 7 and (18) in Lemma 2, we have:

$$\mathbb{E} [f(W^{(\tau)})] - \mathbb{E} [f(W^{(\tau-1)})] \leq 3 \left(\sqrt{2\sigma^2 + 2c^2} + \frac{L}{2} \right) (\eta^2 b^4 c^2 + \eta^2 a^4 c^2 + \eta^4 a^2 b^2 c^4).$$

By applying the inequality above repeatedly, we obtain:

$$\mathbb{E} [f(W^{(\tau)})] \leq \mathbb{E} [f(W^{(0)})] + 3\tau \left(\sqrt{2\sigma^2 + 2c^2} + \frac{L}{2} \right) (\eta^2 b^4 c^2 + \eta^2 a^4 c^2 + \eta^4 a^2 b^2 c^4),$$

which implies $\mathbb{E} [f(W^{(\tau)})]$ is bounded.

Finally, let $\epsilon < \frac{1}{2}$, $\tau \geq \tau_{\text{mix}}(\pi_{\text{min}}\epsilon)$, $T \geq \tau^2$, $\eta \leq \frac{1}{4L\sqrt{T}}$, we obtain:

$$\frac{1}{T} \sum_{t=1}^T \left(\mathbb{E} [\|\nabla_A f(W^{(t)})\|^2] + \mathbb{E} [\|\nabla_B f(W^{(t)})\|^2] \right) \leq \tilde{\mathcal{O}} \left(\epsilon^2 + \frac{L\tau}{\sqrt{T}} + \frac{L\tau}{T} \right).$$

□